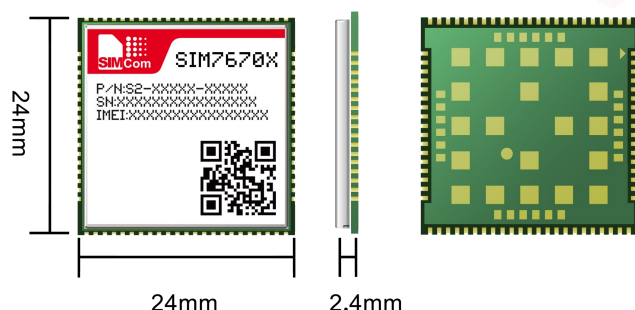


# SIM7670X Series

## SIMCom LTE Cat 1bis Module



## Product Description

SIM7670X series is the LTE Cat 1bis module based on the latest QCX216 chipset that supports wireless communication modes of LTE-FDD/LTE-TDD. It supports maximum 10Mbps downlink rate and 5Mbps uplink rate.

SIM7670X series adopts LCC+LGA form factor and is compatible with A7670 series (Cat1 modules), SIM7000/SIM7070 series (NB/Cat M modules), and SIM800/SIM800F series (2G modules), which enables smooth migration from 2G/NB/Cat M products to LTE Cat 1 products and greatly facilitates more compatible product design for the customer needs.

SIM7670X series supports both multiple built-in network protocols and the drivers for three main operation systems (USB driver for Windows, Linux and Android). The software functions, AT commands are compatible with the A7670X series modules. SIM7670 series also integrates GPS/GLONASS/Galileo/BeiDou Multi-GNSS receiver as well as abundant industrial standard interfaces with powerful expansibility, such as UART, USB, I2C and GPIO, which makes it perfectly suitable for main IOT applications such as telematics, metering, surveillance devices, industrial routers, and remote diagnostics etc.

## Key Benefits

- ◆ Compact size with abundant interfaces and GNSS
- ◆ Abundant software functions: FOTA, SSL, LBS\*
- ◆ High performance Multi-GNSS receiver integrated
- ◆ Form factor is compatible with the A7670/SIM7000/SIM7070 series and 2G SIM800F/SIM800

| Product                        | SIM7670G                                                                                                           | SIM7670NA                                                       | SIM7670E                                                 |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|
| Form Factor                    | LCC+LGA                                                                                                            | LCC+LGA                                                         | LCC+LGA                                                  |
| Dimensions(mm)                 | 24.0*24.0*2.4                                                                                                      | 24.0*24.0*2.4                                                   | 24.0*24.0*2.4                                            |
| Frequency Bands                | LTE-FDD                                                                                                            | B1/B2/B3/B4/B5/B7/B8/B12/B13/B18/B19/B20/B25/B26/B28/B66/B71#   | B1/B3/B5/B7/B8/B20/B28                                   |
|                                | LTE-TDD                                                                                                            | B34/B38/B39/B40/B41                                             | NA                                                       |
|                                | GSM                                                                                                                | NA                                                              | NA                                                       |
| Normal Operation Temperature   | -30°C ~ +75°C                                                                                                      | -30°C ~ +75°C                                                   | -30°C ~ +75°C                                            |
| Extended Operation Temperature | -40°C ~ +85°C                                                                                                      | -40°C ~ +85°C                                                   | -40°C ~ +85°C                                            |
| <b>Electrical Features</b>     |                                                                                                                    |                                                                 |                                                          |
| Supply Voltage(V)              | 3.4~ 4.2                                                                                                           | 3.4~ 4.2                                                        | 3.4~ 4.2                                                 |
| Power Consumption (mA)         | LTE 4.2mA@Idle mode<br>800uA@Sleep mode, DRX=0.64s                                                                 | 4.2mA@Idle mode<br>800uA@Sleep mode, DRX=0.64s                  | 4.2mA@Idle mode<br>800uA@Sleep mode, DRX=0.64s           |
| <b>Data Transfer</b>           |                                                                                                                    |                                                                 |                                                          |
| LTE(Mbps)                      | 10(DL)/5(UL)                                                                                                       | 10(DL)/5(UL)                                                    | 10(DL)/5(UL)                                             |
| <b>Software Features</b>       |                                                                                                                    |                                                                 |                                                          |
| Protocol                       | TCP/IP/IPV4/IPV6/Multi-PDP/FTP/FTPS /HTTP/HTTPS/DNS/MQTT                                                           | TCP/IP/IPV4/IPV6/Multi-PDP/FTP/FTPS /HTTP/HTTPS/DNS/MQTT        | TCP/IP/IPV4/IPV6/Multi-PDP/FTP/FTPS /HTTP/HTTPS/DNS/MQTT |
| TLS                            | •                                                                                                                  | •                                                               | •                                                        |
| File System                    | •                                                                                                                  | •                                                               | •                                                        |
| WIFISCAN                       | •                                                                                                                  | •                                                               | •                                                        |
| LBS                            | o                                                                                                                  | o                                                               | o                                                        |
| FOTA                           | •                                                                                                                  | •                                                               | •                                                        |
| Android RIL                    | Android 5.0/6.0/7.0/8.0/9.0                                                                                        | Android 5.0/6.0/7.0/8.0/9.0                                     | Android 5.0/6.0/7.0/8.0/9.0                              |
| USB Driver                     | Microsoft Windows 7/8/10<br>Linux/Android                                                                          | Microsoft Windows 7/8/10<br>Linux/Android                       | Microsoft Windows 7/8/10<br>Linux/Android                |
| WWAN (RNDIS)                   | •                                                                                                                  | •                                                               | •                                                        |
| VoLTE                          | NA                                                                                                                 | NA                                                              | NA                                                       |
| ECM                            | •                                                                                                                  | •                                                               | •                                                        |
| Embedded AT                    | o                                                                                                                  | o                                                               | o                                                        |
| Firmware Upgrade               | USB/FOTA                                                                                                           | USB/FOTA                                                        | USB/FOTA                                                 |
| <b>Interfaces</b>              |                                                                                                                    |                                                                 |                                                          |
| SIM Card                       | 1.8V/3.0V                                                                                                          | 1.8V/3.0V                                                       | 1.8V/3.0V                                                |
| UART                           | •                                                                                                                  | •                                                               | •                                                        |
| USB                            | •                                                                                                                  | •                                                               | •                                                        |
| ADC                            | •                                                                                                                  | •                                                               | •                                                        |
| PCM                            | •                                                                                                                  | •                                                               | •                                                        |
| GPIO                           | •                                                                                                                  | •                                                               | •                                                        |
| I2C                            | •                                                                                                                  | •                                                               | •                                                        |
| GNSS                           | o                                                                                                                  | o                                                               | o                                                        |
| <b>Certification</b>           |                                                                                                                    |                                                                 |                                                          |
| Certification                  | CE-RED/FCC/TELEC/JATE/<br>GCF*/PTCRB*/RCM*/IC*/NCC*/KC*/<br>CCC*/CTA*/SRRC*/RoHS/REACH/<br>AT&T*/TMOBILE*/VERIZON* | FCC/IC*/GCF*/PTCRB*/<br>RoHS/REACH/AT&T*/<br>T-MOBILE*/VERIZON* | CE-RED/UKCA*/RCM*/<br>RoHS/REACH/DT*/GCF*                |

•: support o/#: optional \* : means under developing/on-going/planning